

第八课预习报告

checking.c

- 作用: 验证输入是否合法并计算指定整数区间的平方和
- 功能:
程序首先要求用户输入一个区间的下限与上限（范围从 -100000000 到 +100000000），验证输入是否为整数且上下限是否合法，然后计算区间内各整数平方的总和并输出结果。
- 演示效果:

```
1 // checking.c -- validating input
2 #include <stdio.h>
3 #include <stdbool.h>
4 // validate that input is an integer
5 long get_long(void);
6 // validate that range limits are valid
7 bool bad_limits(long begin, long end,
8                 long low, long high);
9 // calculate the sum of the squares of the integers
10 // a through b
11 double sum_squares(long a, long b);
12 int main(void)
13 {
14     const long MIN = -100000000L; // lower limit to range
15     const long MAX = +100000000L; // upper limit to range
16     long start; // start of range
17     long stop; // end of range
18     double answer;
19
20     printf("This program computes the sum of the squares of "
21           "integers in a range.\nThe lower bound should not "
22           "be less than -100000000 and\nthe upper bound "
23           "should not be more than +100000000.\nEnter the "
24           "limits (enter 0 for both limits to quit):\n"
25           "lower limit: ");
26     start = get_long();
27     printf("upper limit: ");
28     stop = get_long();
29     while (start != 0 || stop != 0)
```

```

30     {
31         if (bad_limits(start, stop, MIN, MAX))
32             printf("Please try again.\n");
33         else
34         {
35             answer = sum_squares(start, stop);
36             printf("The sum of the squares of the integers ");
37             printf("from %ld to %ld is %g\n",
38                 start, stop, answer);
39         }
40         printf("Enter the limits (enter 0 for both "
41             "limits to quit):\n");
42         printf("lower limit: ");
43         start = get_long();
44         printf("upper limit: ");
45         stop = get_long();
46     }
47     printf("Done.\n");
48
49     return 0;
50 }
51
52 long get_long(void)
53 {
54     long input;
55     char ch;
56
57     while (scanf("%ld", &input) != 1)
58     {
59         while ((ch = getchar()) != '\n')
60             putchar(ch); // dispose of bad input
61         printf(" is not an integer.\nPlease enter an ");
62         printf("integer value, such as 25, -178, or 3: ");
63     }
64
65     return input;
66 }
67
68 double sum_squares(long a, long b)
69 {
70     double total = 0;
71     long i;
72
73     for (i = a; i ≤ b; i++)

```

```

74         total += (double)i * (double)i;
75
76     return total;
77 }
78
79 bool bad_limits(long begin, long end,
80                long low, long high)
81 {
82     bool not_good = false;
83
84     if (begin > end)
85     {
86         printf("%ld isn't smaller than %ld.\n", begin, end);
87         not_good = true;
88     }
89     if (begin < low || end < low)
90     {
91         printf("Values must be %ld or greater.\n", low);
92         not_good = true;
93     }
94     if (begin > high || end > high)
95     {
96         printf("Values must be %ld or less.\n", high);
97         not_good = true;
98     }
99
100     return not_good;
101 }

```

```

1 gcc checking.c -o checking && echo "-3`n3`n0`n0" | ./checking
2 This program computes the sum of the squares of integers in a range.
3 The lower bound should not be less than -10000000 and
4 the upper bound should not be more than +10000000.
5 Enter the limits (enter 0 for both limits to quit):
6 lower limit: upper limit: The sum of the squares of the integers from -3 to 3
  is 28
7 Enter the limits (enter 0 for both limits to quit):
8 lower limit: upper limit: Done.

```

echo_eof.c

- 作用: 将标准输入的内容原样输出到标准输出直到遇到文件结束符 (EOF)

- 功能:
程序读取输入的每个字符并立即输出，常用于测试输入输出流或作为数据透传工具。
- 演示效果:

```
1  /* echo_eof.c -- repeats input to end of file */
2  #include <stdio.h>
3  int main(void)
4  {
5      int ch;
6
7      while ((ch = getchar()) != EOF)
8          putchar(ch);
9
10     return 0;
11 }
```

```
1  gcc echo_eof.c -o echo_eof && echo "Hello World!" | ./echo_eof
2  Hello World!
```

echo.c

- 作用: 将输入的字符逐一输出，直到遇到字符 '#' 停止程序。
- 功能:
程序通过 `getchar()` 从标准输入流读取字符，直到检测到终止字符 `#` 为止，将之前的所有字符通过 `putchar()` 输出到标准输出。
- 演示效果:

```
1  /* echo.c -- repeats input */
2  #include <stdio.h>
3  int main(void)
4  {
5      char ch;
6
7      while ((ch = getchar()) != '#')
8          putchar(ch);
9
10     return 0;
11 }
```

```
1 gcc echo.c -o echo && echo "Hello#World" | ./echo
2 Hello
```

file_eof.c

- 作用: 从指定文件中读取并显示内容到标准输出。
- 功能:
程序请求用户输入文件名, 尝试读取并输出该文件的全部内容。如果文件无法打开会给出提示并退出。
- 演示效果:

```
1 // file_eof.c --open a file and display it
2 #include <stdio.h>
3 #include <stdlib.h> // for exit()
4 int main()
5 {
6     int ch;
7     FILE * fp;
8     char fname[50]; // to hold the file name
9
10    printf("Enter the name of the file: ");
11    scanf("%s", fname);
12    fp = fopen(fname, "r"); // open file for reading
13    if (fp == NULL) // attempt failed
14    {
15        printf("Failed to open file. Bye\n");
16        exit(1); // quit program
17    }
18    // getc(fp) gets a character from the open file
19    while ((ch = getc(fp)) != EOF)
20        putchar(ch);
21    fclose(fp); // close the file
22
23    return 0;
24 }
```

```
1 echo "Hello File" > test.txt && gcc file_eof.c -o file_eof && echo "test.txt" |
./file_eof
2 Enter the name of the file: Hello File
```

guess.c

- 作用: 通过不断尝试从 1 到 100 的整数来猜测用户的数字。
- 功能:
使用简单的线性递增方式, 每次询问用户当前猜测的数字是否正确, 直到用户输入 'y' 结束。用户输入 'n' 表示错误, 程序会继续增加猜测值并重复提问。
- 演示效果:

```
1  /* guess.c -- an inefficient and faulty number-guesser */
2  #include <stdio.h>
3  int main(void)
4  {
5      int guess = 1;
6
7      printf("Pick an integer from 1 to 100. I will try to guess ");
8      printf("it.\nRespond with a y if my guess is right and with");
9      printf("\nan n if it is wrong.\n");
10     printf("Uh...is your number %d?\n", guess);
11     while (getchar() != 'y') /* get response, compare to y */
12         printf("Well, then, is it %d?\n", ++guess);
13     printf("I knew I could do it!\n");
14
15     return 0;
16 }
```

```
1  gcc guess.c -o guess && echo
   "n`nn`nn`nn`nn`nn`nn`nn`nn`nn`nn`nn`nn`nn`nn`nn`nn`nn`nn`nn`ny" | ./guess
2  Pick an integer from 1 to 100. I will try to guess it.
3  Respond with a y if my guess is right and with
4  an n if it is wrong.
5  Uh...is your number 1?
6  Well, then, is it 2?
7  Well, then, is it 3?
8  Well, then, is it 4?
9  Well, then, is it 5?
10 Well, then, is it 6?
11 Well, then, is it 7?
12 Well, then, is it 8?
13 Well, then, is it 9?
14 Well, then, is it 10?
15 Well, then, is it 11?
```

```
16 Well, then, is it 12?
17 Well, then, is it 13?
18 Well, then, is it 14?
19 Well, then, is it 15?
20 Well, then, is it 16?
21 Well, then, is it 17?
22 Well, then, is it 18?
23 Well, then, is it 19?
24 Well, then, is it 20?
25 Well, then, is it 21?
26 Well, then, is it 22?
27 Well, then, is it 23?
28 Well, then, is it 24?
29 Well, then, is it 25?
30 Well, then, is it 26?
31 Well, then, is it 27?
32 Well, then, is it 28?
33 Well, then, is it 29?
34 Well, then, is it 30?
35 Well, then, is it 31?
36 Well, then, is it 32?
37 Well, then, is it 33?
38 Well, then, is it 34?
39 Well, then, is it 35?
40 Well, then, is it 36?
41 Well, then, is it 37?
42 Well, then, is it 38?
43 Well, then, is it 39?
44 Well, then, is it 40?
45 Well, then, is it 41?
46 I knew I could do it!
```

showchar1.c

- 作用: 读取一个字符和两个整数, 然后将该字符格式化输出为指定行数和列数的矩阵。
- 功能:
用户输入一个字符和两个整数, 分别表示输出字符以及字符矩阵的行数和列数。程序调用 `display()` 函数按指定格式打印字符阵列。
(此程序存在逻辑问题: 输入后换行符处理不完善可能导致混乱。)
- 演示效果:

```
1 /* showchar1.c -- program with a BIG I/O problem */
```

```

2  #include <stdio.h>
3  void display(char cr, int lines, int width);
4  int main(void)
5  {
6      int ch;          /* character to be printed */
7      int rows, cols;   /* number of rows and columns */
8      printf("Enter a character and two integers:\n");
9      while ((ch = getchar()) != '\n')
10     {
11         scanf("%d %d", &rows, &cols);
12         display(ch, rows, cols);
13         printf("Enter another character and two integers:\n");
14         printf("Enter a newline to quit.\n");
15     }
16     printf("Bye.\n");
17
18     return 0;
19 }
20
21 void display(char cr, int lines, int width)
22 {
23     int row, col;
24
25     for (row = 1; row ≤ lines; row++)
26     {
27         for (col = 1; col ≤ width; col++)
28             putchar(cr);
29         putchar('\n'); /* end line and start a new one */
30     }
31 }

```

```

1  gcc showchar1.c -o showchar1 && echo "@`n2 4`n`n" | ./showchar1
2  Enter a character and two integers:
3  @@@@
4  @@@@
5  Enter another character and two integers;
6  Enter a newline to quit.
7  Bye.

```

showchar2.c

- 作用: 读取一个字符和两个整数, 并按指定行数和列数输出该字符矩阵。

- 功能:

用户输入一个字符和两个整数，程序会调用 `display()` 函数按行列格式输出字符阵列。相比 `showchar1.c`，该程序在输入处理上更健壮，能正确跳过多余的输入。

- 演示效果:

```
1  /* showchar2.c -- prints characters in rows and columns */
2  #include <stdio.h>
3  void display(char cr, int lines, int width);
4  int main(void)
5  {
6      int ch;          /* character to be printed */
7      int rows, cols;  /* number of rows and columns */
8
9      printf("Enter a character and two integers:\n");
10     while ((ch = getchar()) != '\n')
11     {
12         if (scanf("%d %d",&rows, &cols) != 2)
13             break;
14         display(ch, rows, cols);
15         while (getchar() != '\n')
16             continue;
17         printf("Enter another character and two integers;\n");
18         printf("Enter a newline to quit.\n");
19     }
20     printf("Bye.\n");
21
22     return 0;
23 }
24
25 void display(char cr, int lines, int width)
26 {
27     int row, col;
28
29     for (row = 1; row ≤ lines; row++)
30     {
31         for (col = 1; col ≤ width; col++)
32             putchar(cr);
33         putchar('\n'); /* end line and start a new one */
34     }
35 }
```

```
1 gcc showchar2.c -o showchar2 && echo "@`n2 3`n`n" | ./showchar2
2 Enter a character and two integers:
3 @@@
4 @@@
5 Enter another character and two integers;
6 Enter a newline to quit.
7 Bye.
```

menuette.c

- 作用: 提供一个简单的菜单交互程序, 允许用户选择不同功能, 如显示建议、发出提示音或进行计数。
- 功能:
 - 程序循环显示菜单, 用户可选择:
 - a: 显示“买低卖高”的建议文本
 - b: 发出 ANSI 铃声
 - c: 输入一个整数并按顺序从 1 输出到该整数
 - q: 退出程序
- 演示效果:

```
1  /* menuette.c -- menu techniques */
2  #include <stdio.h>
3  char get_choice(void);
4  char get_first(void);
5  int get_int(void);
6  void count(void);
7  int main(void)
8  {
9      int choice;
10     void count(void);
11
12     while ( (choice = get_choice()) != 'q')
13     {
14         switch (choice)
15         {
16             case 'a' : printf("Buy low, sell high.\n");
17                         break;
18             case 'b' : putchar('\a'); /* ANSI */
19                         break;
```

```

20         case 'c' : count();
21         break;
22         default : printf("Program error!\n");
23         break;
24     }
25 }
26 printf("Bye.\n");
27
28 return 0;
29 }
30
31 void count(void)
32 {
33     int n,i;
34
35     printf("Count how far? Enter an integer:\n");
36     n = get_int();
37     for (i = 1; i ≤ n; i++)
38         printf("%d\n", i);
39     while ( getchar() ≠ '\n')
40         continue;
41 }
42
43 char get_choice(void)
44 {
45     int ch;
46
47     printf("Enter the letter of your choice:\n");
48     printf("a. advice          b. bell\n");
49     printf("c. count           q. quit\n");
50     ch = get_first();
51     while ( (ch < 'a' || ch > 'c') && ch ≠ 'q')
52     {
53         printf("Please respond with a, b, c, or q.\n");
54         ch = get_first();
55     }
56
57     return ch;
58 }
59
60 char get_first(void)
61 {
62     int ch;
63

```

```

64     ch = getchar();
65     while (getchar() != '\n')
66         continue;
67
68     return ch;
69 }
70
71 int get_int(void)
72 {
73     int input;
74     char ch;
75
76     while (scanf("%d", &input) != 1)
77     {
78         while ((ch = getchar()) != '\n')
79             putchar(ch); // dispose of bad input
80         printf(" is not an integer.\nPlease enter an ");
81         printf("integer value, such as 25, -178, or 3: ");
82     }
83
84     return input;
85 }

```

```

1 gcc menuette.c -o menuette && echo "c`n5`nq" | ./menuette
2 Enter the letter of your choice:
3 a. advice          b. bell
4 c. count           q. quit
5 Count how far? Enter an integer:
6 1
7 2
8 3
9 4
10 5
11 Enter the letter of your choice:
12 a. advice          b. bell
13 c. count           q. quit
14 Bye.

```

newguess.c

- 作用: 改进的数字猜测程序, 可以处理用户输入非 'y' 或 'n' 的情况并给出提示。

- 功能:

与 `guess.c` 类似, 通过逐步增加猜测值尝试猜测用户的数字。新增了对非法输入的处理, 确保用户必须输入 'y' 或 'n'。

- 演示效果:

```
1  /* newguess.c -- an inefficient but improved number-guesser */
2  #include <stdio.h>
3  int main(void)
4  {
5      int guess = 1;
6      char response;
7
8      printf("Pick an integer from 1 to 100. I will try to guess ");
9      printf("it.\nRespond with a y if my guess is right and with");
10     printf("\nan n if it is wrong.\n");
11     printf("Uh...is your number %d?\n", guess);
12     while ((response = getchar()) != 'y')    /* get response */
13     {
14         if (response == 'n')
15             printf("Well, then, is it %d?\n", ++guess);
16         else
17             printf("Sorry, I understand only y or n.\n");
18         while (getchar() != '\n')
19             continue;    /* skip rest of input line */
20     }
21     printf("I knew I could do it!\n");
22
23     return 0;
24 }
```

```
1  gcc newguess.c -o newguess && echo "n`nn`nx`nn`ny" | ./newguess
2  Pick an integer from 1 to 100. I will try to guess it.
3  Respond with a y if my guess is right and with
4  an n if it is wrong.
5  Uh...is your number 1?
6  Well, then, is it 2?
7  Well, then, is it 3?
8  Sorry, I understand only y or n.
9  Well, then, is it 4?
10 I knew I could do it!
```